

MACAE, Brazil

WMO: 868910

Lat: 22.343S

Lon: 41.766W

Elev: 2

StdP: 101.30

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	14.9	11.2	8.3	18.9	12.5	9.1	19.1	8.4	24.0	7.6	24.5	1.6	320	0.301

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.8	36.8	26.1	35.0	25.5	33.4	25.1	27.6	32.8	27.0	31.5	26.6	30.8	4.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.7	29.1	26.0	21.4	28.9	25.2	20.4	28.1	87.7	33.1	85.2	31.8	83.1	30.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	11.0	39.9	1.1	2.0	10.2	41.3	9.6	42.5	9.0	43.6	8.2	45.0 (4)		
(5)			10.6	29.3	1.1	1.2	9.8	30.2	9.1	30.9	8.5	31.6	7.6	32.5 (5)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	27.6	28.1	27.1	25.4	23.2	22.2	22.0	22.2	23.5	24.3	25.1	26.6	(6)
(7)		DBStd	3.21	2.40	1.96	2.17	2.13	2.13	2.26	2.46	2.49	2.84	2.77	2.72	2.43	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	4	0	0	0	0	0	1	2	1	1	0	0	0	(9)
(10)		CDD10.0	5384	546	508	531	462	409	366	372	378	404	442	453	515	(10)
(11)		CDD18.3	2347	287	274	272	212	151	116	115	121	154	184	203	257	(11)
(12)		CDH23.3	17464	2634	2634	2377	1579	886	641	644	693	871	1043	1362	2099	(12)
(13)		CDH26.7	6258	1071	1103	869	483	233	183	208	243	329	340	441	755	(13)
(14)	Wind	WSAvg	2.9	3.2	3.1	2.5	2.4	2.3	2.4	2.7	3.1	3.4	3.4	3.2	3.0	(14)
(15)	Precipitation	PrecAvg	1362	194	123	148	113	64	40	46	98	118	185	200	(15)	
(16)		PrecMax	1664	352	391	377	209	90	103	118	95	261	208	310	258	(16)
(17)		PrecMin	1027	33	8	33	48	15	12	9	15	18	53	104	128	(17)
(18)		PrecStd	177	78	87	80	52	21	23	30	23	58	47	53	47	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	38.5	37.9	35.1	33.2	33.1	33.6	35.8	36.1	35.9	36.1	38.2	(19)
MCWB			27.2	26.7	27.0	24.9	23.7	22.9	22.2	22.3	24.1	24.7	25.6	27.2	(20)	
2%		DB	35.2	35.8	34.3	32.5	31.1	30.8	31.1	31.8	32.9	32.2	33.1	34.8	(21)	
		MCWB	26.4	25.9	26.2	25.0	24.0	22.4	21.8	21.7	23.5	24.1	25.3	26.0	(22)	
5%		DB	33.2	33.8	32.2	30.9	29.1	28.8	29.0	29.1	30.2	30.1	31.0	32.2	(23)	
		MCWB	25.7	25.8	25.9	25.0	23.3	22.1	21.2	21.6	22.5	23.4	25.1	25.4	(24)	
10%		DB	31.9	32.0	30.9	29.1	27.5	26.9	27.0	27.0	28.0	28.1	29.0	30.7	(25)	
		MCWB	25.3	25.3	25.8	24.5	23.0	21.7	21.3	21.2	21.8	23.0	24.4	25.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.5	28.0	28.4	27.1	26.4	24.5	24.0	25.2	26.4	27.0	27.8	(27)	
MCDWB			34.1	34.9	33.7	30.3	28.8	29.1	28.8	30.0	32.4	32.3	31.8	33.3	(28)	
2%		WB	27.5	27.1	27.3	26.4	25.2	23.7	23.1	23.0	24.3	25.2	26.3	27.0	(29)	
		MCDWB	32.4	32.2	32.0	29.9	28.3	27.8	27.7	28.3	30.0	30.1	30.7	31.4	(30)	
5%		WB	26.7	26.5	26.7	25.8	24.5	23.0	22.5	22.3	23.5	24.5	25.7	26.3	(31)	
		MCDWB	30.8	31.1	31.0	29.4	27.7	26.7	26.8	27.0	28.2	28.5	29.4	30.4	(32)	
10%		WB	26.2	26.1	26.2	25.1	23.7	22.3	21.7	21.7	22.7	23.8	25.1	25.8	(33)	
		MCDWB	29.9	30.3	30.0	28.4	26.8	25.8	25.6	25.8	26.4	27.0	28.0	29.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	8.7	8.9	9.2	10.3	10.3	9.5	8.1	7.8	8.2	(35)	
MCDBR			12.1	12.3	11.3	11.1	11.9	12.8	13.5	14.4	13.6	11.6	11.3	11.5	(36)	
MCWBR			4.2	3.9	4.5	4.8	5.3	5.8	5.3	5.7	4.8	4.4	4.1	4.2	(37)	
5% WB		MCDBR	10.3	10.2	9.9	9.7	9.8	11.4	11.6	11.9	11.6	10.0	9.3	9.6	(38)	
		MCWBR	4.2	4.1	4.4	4.6	5.2	5.9	5.9	5.7	4.7	4.2	3.9	4.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.407	0.399	0.402	0.383	0.383	0.375	0.370	0.408	0.476	0.464	0.438	0.422	(40)	
taud		2.480	2.489	2.491	2.529	2.497	2.514	2.509	2.382	2.167	2.257	2.363	2.429	(41)		
Ebn at Noon		939	933	904	879	837	824	843	841	822	863	904	926	(42)		
Edh at Noon		118	115	110	98	94	88	91	112	150	143	132	124	(43)		
(44)	All-Sky Solar Radiation	RadAvg	6.27	6.43	5.48	4.76	3.84	3.59	3.70	4.42	4.75	5.30	5.35	5.93	(44)	
(45)		RadStd	0.94	0.58	0.43	0.28	0.16	0.22	0.27	0.29	0.51	0.58	0.48	0.66	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	-1.69	N/A	-1.40	-1.66	N/A	N/A	N/A	N/A

Nomenclature: See separate page